

ACADEMIC PORTFOLIO

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Software Engineering → Quantum Information

How can imperfect quantum resources be transformed into reliable computational advantages?

Quantum Information · Quantum Communication
Quantum Cryptography · Quantum Networks
Quantum Games · Quantum Machine Intelligence

Luoyang Normal University | Fall 2027 Applicant



On Grover-based quantum signature schemes

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Abstract: Quantum computation, with tools including Grover's algorithm, quantum walks, and quantum teleportation, plays an important role in quantum signature designs. Although each design offers signature functionality, they sometimes come at the cost of security. This paper first reviews a Grover-based scheme and shows how man-in-the-middle attacks enable Alice's discovery and Bob's forgery, which motivates our proposal of four inspection points to identify such vulnerabilities. We then propose an arbitrated quantum signature scheme using a strengthened quantum one-time pad to reduce the risk of forgery, while quantum teleportation enables channel reuse and prevents eavesdropping. Moreover, the proposed scheme successfully passes all four inspection points and achieves undetectability and non-repudiation, leveraging the inherent security guarantees of quantum physics. Finally, its practicality is further validated by resource consumption analysis and simulation results.

Keywords: Quantum computation; Grover algorithm; Quantum signature; Security analysis

1. Introduction

In the field of quantum computation and communication, Grover's algorithm, quantum teleportation and quantum walk have inspired a variety of cryptographic constructions far beyond their original application scenarios [1–7]. Leveraging quantum superposition, phase manipulation and interference, these algorithms-driven schemes feature concise and flexible architecture, and exhibit inherent quantum resilience compared with conventional digital signature schemes. Among them quantum primitives, Grover's algorithm, as a fundamental building block in quantum computation and complexity theory, is particularly intriguing and conceptually appealing, offering a novel perspective for the design and construction of quantum signature schemes.

Among quantum signature schemes, arbitrated quantum signature (AQS) schemes introduce a trusted or semi-trusted arbitrator to assist verification and dispute resolution [8–10]. Since the early scheme of Zeng and Ketel [8], a range of AQS constructions have been proposed using entanglement, Bell states, or quantum one-time pad (QOTV) encryption [9, 11–13]. At the same time, the design of secure AQS protocols has proved difficult. A number of schemes built from seemingly sound quantum primitives were later shown to admit forgery, repudiation, or impersonation attacks [14–16]. These results suggest that the security of quantum signature schemes depends not only on the local soundness of the primitives employed, but also on the way those primitives are combined at the protocol level.

Version 141020 submitted to Journal Not Specified

https://doi.org/10.3390/1410000

Weak-link entanglement-assisted no-signaling quantum communication protocols: Analytical framework and cross-platform verifiability

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Abstract: We introduce a weak-link entanglement-assisted no-signaling quantum communication protocol that bridges cooperative and fault-tolerant tasks in weak-link quantum networks. Analytical first-order correction for AND and OR communication functions are derived from an expansion of the joint success probability, establishing a tunable trade-off between cooperative gain and individual tolerance. A unified tensor-negating model connects anisotropic noise—Werner, depolarizing, phase, and amplitude damping—to effective link vulnerabilities while preserving central nonlocality-based under isotropic conditions. Monte Carlo and Kraus-operator simulations confirm the protocol's resilience across 2–5 node networks and its scalability under mixed noise environments. Feasibility analyses for photonic, trapped-ion, and superconducting architectures show that current device parameters fully cover the weak-link operational regime. This work establishes the first general framework testing weak entanglement links as functional communication resources under no-signaling constraints, providing a platform-independent, experimentally verifiable foundation for scalable quantum networking.

1. Introduction

Constructing scalable and experimentally verifiable quantum communication networks under realistic noise is pushing quantum information processing beyond “toy” demonstrations (e.g., QKD and quantum teleportation) toward multi-node, multi-layer, and multi-task architectures. Foundational works have shown that allowing quantum operations and entanglement resources enables strategies and cooperative capabilities that surpass classical limits (e.g., quantum strategies in the quantum Prisoner's Dilemma [1, 2, 7]). Under the no-signaling constraint, pseudo-telepathy further demonstrates that entanglement can generate non-classical consistency even when no information exchange is allowed [2, 7].

Meanwhile, network-level connectivity and resource topology—including bilateral configurations, full-network undirected, and starlike or ringlike topologies—nonclassically—have become key perspectives for understanding how multi-source entanglement converts into system-level performance [2, 7, 7].

The practical bottleneck is not whether entanglement “exists” but the objectively ubiquitous weak links in real networks, whose viability/fidelity is significantly lower than that of backbone links. Long-haul experiments in metropolitan fiber networks, heterogeneous satellite-ground interfaces, and imperfect coupling in multi-core superconducting prototypes. Conventional approaches often treat weak links as passive bottlenecks to be bypassed via heavy distillation or relay, which becomes resource-intensive under NNN constraints. Another viewpoint models the network as a percolation system, describing large-scale connectivity through phase transition or percolation [7]. However, such weak links largely

remain at the level of feasibility criteria. A fully operational protocol model is still lacking—one that maps weak-link connectivity directly to task-level performance and tunable engineering parameters.



FIG. 1: Schematic diagram of the overall framework of the agreement. Used to display conceptual models.

This work fills exactly this gap: we propose and systematize a class of weak-link entanglement-assisted no-signaling quantum communication protocols. The core idea (Fig. 1) is to partition the network nodes into several cooperating groups. Within each group, strong co-incident provides stable links and internal consistency; between groups, only a sparse cluster of weak entanglement links is needed. All parties perform strictly local measurements with no classical or quantum communication, fully respecting the no-signaling condition. The protocol outputs are not framed as sequentially verifiable but as two task success probabilities:

- **AND-type:** all cooperating groups succeed simultaneously, characterizing high-consistency / strong-coordination tasks;
- **OR-type:** at least one group succeeds, characterizing redundancy / fault-tolerant tasks.

The operational effect appears as a first-order network correction:

$$\Delta = \sum_{i,j} \sum_{k,l} h_{ij} h_{kl} p_{ij} p_{kl}$$

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Error-Corrected Quantum-Huffman-Inspired Probabilistic Source Decoding with Group Ancilla and Posterior Correction

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We study a finite-resource probabilistic source-decoding model inspired by quantum Huffman coding, but not a strictly infinite quantum Huffman code. Huffman lengths are interpreted as finite-dimensional support budgets rather than transmitted coherent variable-length quantum codewords, producing a source-decoding nonorthogonal ensemble in a common Hilbert space. A group ancilla transfers inter-group undecidability into orthogonal local information, leaving middle conditional PVTM tasks. Resource-constrained mappings then decide whether success gains pass for group-level and reliability overhead. We distinguish representative benchmarking from a favorable-regime parameter sweep; the former shows that resource advantage is not purely under conservative overhead, whereas the latter shows that positive margin energy structured low-overhead, high-support-swing regimes. Exact SDP probes verify selected finite instances, including a controlled $M = 8$ probe, while large- M ones are treated as PVTM diagnostics. QEC and MAP are ancillary reliability and posterior-decision layers, not the central coding core.

1. INTRODUCTION

Classical Huffman coding converts a nonuniform source distribution into prefix lengths L_i satisfying the Kraft inequality [23].

$$\sum_i 2^{-L_i} \leq 1. \quad (1)$$

with average length $L = \sum_i p_i L_i$. A direct quantum analogue is subtle because coherent superpositions of different codeword lengths make termination and length measurement nontrivial [8,12]. The present work therefore does not construct a strictly infinite quantum Huffman code. Instead, it uses classical Huffman lengths as support-budget parameters in a common finite-dimensional Hilbert space and studies probabilistic decoding of the resulting nonorthogonal states.

The difficulty is not merely technical bookkeeping. In a coherent variable-length code, a quantum source may occupy a superposition of different length sectors. A decoder that tries to identify a codeword boundary or a prefix termination thus runs into local information and thereby distorts the encoded state. Schumacher compensation and its refinements purely analyze quantum and finite-resource perspectives on quantum source coding [8,12,13,23]. They do not by themselves provide a finite-sized, real-time Huffman tree with coherent prefix parsing. These observations motivate a more model-independent question: can the probability structure that makes classical Huffman coding useful still be used to shape a quantum ensemble, even if the final decoding is probabilistic rather than lossless?

Our answer is to separate “Huffman structure” from “variable-length transmission.” The length L_i is treated as a support-budget parameter, so that 2^{-L_i} determine a relative share of a common ambient Hilbert space. High-probability symbols receive more amplitude phase-shaping freedom, while all symbol states remain embedded in the same Hilbert space for discrimination. This avoids the coherence problem of measuring length, but preserves the source-dependent geometry induced by the classical Huffman tree.

The central contribution is the combination of Huffman-inspired support budgeting and group-confined state discrimination under an explicit resource-efficiency cost. Huffman lengths determine how much support each source symbol may occupy. The induced support budget controls state overlap. An orthogonal group ancilla separates inter-group alternative codewords and leaves middle conditional PVTM tasks inside each group. QEC rates only through an effective logical failure mode, and MAP correction is a block-level deterministic post-processing rule. The final question is whether the success gain compensates the group-level and physical-scale overhead.

The words “error-corrected” and “posterior correction” in the title should therefore be read in this limited sense. QEC is an effective reliability interface, and MAP is a minimal posterior decision post-processing. Neither layer is the main coding novelty; a new quantum Huffman mechanism, or a hardware-level fault-tolerant implementation, Compared with strict variable-length quantum coding,

A learnable unified framework for weak-link multi-group quantum games

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Abstract

This paper presents a learnable framework for weak-link multi-group quantum games (MGQG) from the perspective of quantum machine intelligence. In the multi-link layer, cooperative groups are represented as vertices of a weak-link interaction graph, and the leading inter-group contribution is encoded by the size of the game matrix. This interaction yields optimal joint first-order advantage criteria, spectral upper bounds, reliability estimates, and a task-dependent interpretation of noise. In the learnable layer, the discrete analytical structure is relaxed into differentiable soft signs, tunable bias terms, and adaptive variables, enabling a hybrid quantum-classical optimization pipeline over spectral and regularization conditions. In the computational layer, surrogate learning, recurrent neural networks, and gradient descent are used to counter the behavior of the proposed framework. The results provide surrogate-level evidence that learned quantum game structures can reliably coordinate with analytical sign rules while allowing tunable adaptation across topology, task structure, and noise conditions. The paper distinguishes theoretical first-order analysis from heuristic surrogate objectives and hardware-ready game quantities, providing a structured route from analytical quantum game models to tunable quantum machine intelligence workflows.

Research Identity

A problem-driven trajectory from secure protocols to adaptive quantum networks

CENTRAL QUESTION

How can imperfect quantum resources be transformed into reliable computational advantages?

My research treats noise, limited entanglement, protocol exposure, and finite computational budgets not only as limitations, but as structures that can be modeled, allocated, and optimized.

RESEARCH PREPARATION

- B.Eng. in Software Engineering; academic rank in the top 10%.
- Supervisor-led reading group in quantum information, computing, cryptography, and communication.
- Python, Qiskit, PennyLane, numerical simulation, optimization, and LaTeX research workflows.
- Experience converting analytical models into reproducible code, figures, tables, and manuscripts.

RESEARCH TRAJECTORY

01	Protocol security How does system structure determine whether quantum primitives remain secure?
02	Finite-resource decoding When does a raw performance gain survive explicit resource accounting?
03	Weak-link communication Can noisy and heterogeneous entanglement links become functional resources?
04	Adaptive quantum games Can distributed quantum networks learn task-aware decision policies?
05	Future Quantum Internet How can communication, computation, security, and learning be optimized together?

03. Structural Security of Quantum Signatures

Quantum primitives are secure only when the full protocol architecture is secure.

RESEARCH PROBLEM Grover operations, teleportation, and QKD may each be legitimate primitives, yet their composition can expose verification structure, keys, and message origin.	CORE APPROACH Analyze complete information flow; abstract reusable inspection points; redesign the architecture with arbitration, strengthened QOTP, and teleportation.	MY CONTRIBUTION Organized mathematical framework and notation; supported four reusable inspection points; implemented Qiskit verification; contributed to the strengthened construction and manuscript.
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STATUS Second author · Revised and resubmitted

(AE1) In Step (AT4), Eve intercepts the first sequence of particles of the message qubits $U_{S_{AT}}|M\rangle$ and the message that Alice tells TC the locations of half the decoy qubits.

(AE2) In Step (AT5), Eve performs measurement on the decoy qubits using Z or X basis, and impersonates TC to tell Alice her measurement bases and measurement outcomes.

(AE3) In Step (AT6), Eve intercepts the other sequence of the particles that Alice sends to TC and the message that Alice tells TC the positions of decoy qubits and the types of unitary operations.

(AE4) In Step (AT8), Eve makes the unitary operator $U_V = 2|M\rangle\langle M| - I$ based on the known message. Then, she implements U_V on the received message to recover the shared key as $U_V U_{S_{AT}}|M\rangle = \pm|K_{AT}\rangle$. Thereby, Eve can obtain the shared key K_{AT} successfully.

In the Process between Eve and TC, Eve performs the following two steps to pass TC's verify by posing as Alice.

(ET1) Eve pretends to be Alice to announce forged message m' and makes its signature $U_{S_{AT}}|M'\rangle$ with K_{AT} .

(ET2) Eve impersonates Alice to communicate with TC by performing Steps (AT4)-(AT8).

Finally, there is nothing Eve need to do in the process between TC and Bob.

By performing the MITM attack above, $U_{S_{AT}}|M'\rangle$ as the signature of the forged message m' can pass TC's verifying. Therefore, Eve can forge Alice's signature for any message successfully.

Remark 1. Similarly, to forge Alice's signature in YKLY'2015 [4], Eve can perform an MITM attack between TC and Bob: Eve intercepts the message qubits $U_{S_{TB}}|M\rangle$ to obtain the shared key K_{TB} , pretends to be Alice to announce forged message m' , makes the forged signature $U_{S_{TB}}|M'\rangle$ with K_{TB} , and sends $U_{S_{TB}}|M'\rangle$ to Bob as TC in YKLY'2015. In particular, if the attacker is Alice, her disavowal can be achieved.

SELECTED EVIDENCE

- MITM analysis identifies key recovery, forgery, and repudiation pathways.
- Four inspection points turn scheme-specific attacks into reusable protocol checks.
- The redesigned construction separates verification authority and strengthens signer-message binding.

NEXT QUESTION Security analysis showed that reliable advantage depends on system-level resource roles. This led to the question of how limited quantum resources should be allocated and evaluated.

04. Resource-Efficient Probabilistic Quantum Decoding

A performance gain is meaningful only when it survives the cost of ancilla, redundancy, and decoding.

RESEARCH PROBLEM

Can classical source statistics guide finite-dimensional quantum state design without claiming a strictly lossless coherent quantum Huffman code?

CORE APPROACH

Reinterpret Huffman lengths as support budgets; construct source-shaped nonorthogonal ensembles; use group ancilla and conditional POVMs; evaluate QEC and MAP under explicit resource-normalized margins.

MY CONTRIBUTION

Developed the support-budget construction, group-conditioned discrimination, analytical conditions, SDP certificate logic, scalable simulations, and sensitivity analysis.

STATUS First author · Manuscript in preparation

$ \Phi_i\rangle$	Payload state supported on S_i .
Γ_{ij}	Payload Gram-matrix entry, $\Gamma_{ij} = \langle \Phi_i \Phi_j \rangle$.
$g(i)$	Group index of symbol s_i .
$\mathcal{G}_g$ or G_g	The g -th group of source symbols.
G	Number of groups.
$ a_g\rangle$	Orthogonal group-ancilla label.
$ \Psi_i\rangle$	Grouped state $ \Phi_i\rangle \otimes a_{g(i)}\rangle$.
$\mathcal{A}_{\text{Huff}}$	Huffman-induced ambiguity budget.
$\mathcal{A}_i$	Symbol-level ambiguity contribution.
$\mathcal{A}_{\text{global}}$	Total source-weighted overlap ambiguity before grouping.
$\mathcal{A}_{\text{in}}$	Ambiguity remaining inside conditional group tasks.
$\mathcal{A}_{\text{out}}$	Ambiguity removed by the group label, $\mathcal{A}_{\text{out}} = \mathcal{A}_{\text{global}} - \mathcal{A}_{\text{in}}$.
P_{group}	Optimal group-conditioned success probability.
P_{full}	Effective full success after QEC reliability and fallback.
P_{base}	Generic baseline success probability.
$P_{\text{base,opt}}$	Numerical exact or representative baseline success.
q_L	Effective logical failure probability after recovery.
P_{fallback}	Success probability of the fallback decision rule.
C_A	Group-label cost, $C_A = \log_2 G$.
C_{QEC}	Modeled QEC redundancy cost.
C_{total}	Total modeled resource cost.
η	Resource-normalized efficiency, $P_{\text{full}}/C_{\text{total}}$.
M_{res}	Resource margin from Eq. (34).
M_{cert}	SDP-certified normalized margin from Eq. (48).

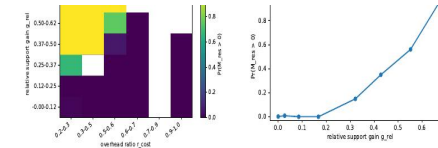
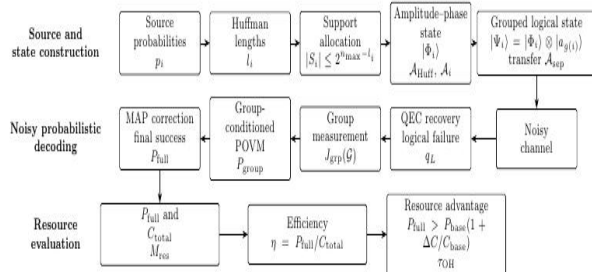


FIG. 6. Favorable-regime sweep, not the default benchmark. The heatmap reports the binned positive-margin fraction for no-QEC PGM-diagnostic rows as a function of relative support gain and overhead ratio. Positive regions appear where support gain is large enough and modeled overhead is low enough; this maps nonempty structured regimes, not universal advantage.

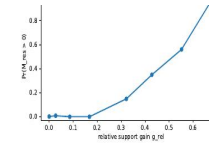


FIG. 7. Favorable-regime sweep probability versus relative support-budget gain q_{res} . Bars are binned PGM diagnostics with bin counts annotated in the generated figure and CSV. Positive margins become more likely at larger support savings, but this statistic is separate from the conservative default benchmark and is not an SDP certificate.

favorable_regime_sweep.csv. This sweep intentionally explores design degrees of freedom such as source skew, support-overlap shift, payload-generation mode, grouping strategy, group number, QEC overhead, and weighted overhead parameters. Its positive fraction is therefore not a typical-case success probability. It is a design-space map used to test whether positive-margin regions are nonempty and structurally interpretable.

The sweep contains 56160 PGM-labeled diagnostic rows generated by the current support-budget payload pipeline. The all-row positive fraction is $6828/56160 = 0.122$, the no-QEC subset has $4812/18720 = 0.257$ positive rows, and the weighted-cost margin has positive fraction 0.347. Positive no-QEC rows have larger mean support gain and lower overhead than non-positive rows: mean $G_{\text{comp}} = 2.52$ versus 0.56, mean $\beta_{\text{rel}} = 0.577$ versus 0.182, mean $r_{\text{exp}} = 0.834$ versus 0.785, and mean $r_{\text{out}} = 0.356$ versus 0.565. Thus positive margins concentrate where support-budget saving, ambiguity transfer and low overhead are resonant simultaneously.

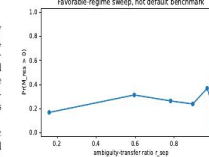


FIG. 8. Favorable-regime sweep probability versus ambiguity-transfer ratio r_{exp} . The binned PGM diagnostic is not monotone enough to support a threshold law: ambiguity transfer helps, but positive margins also require support saving and low modeled overhead.

D. Exact certificate statistics

SELECTED EVIDENCE

- Default benchmarks show that reliability does not automatically imply resource advantage.
- Favorable-regime sweeps identify low-overhead, high-support-saving regions.
- Exact finite-instance SDP probes provide sufficient certificates rather than universal claims.

NEXT QUESTION The framework suggested that imperfect resources can be reshaped, not merely compensated, motivating weak entanglement as a task-aware network resource.

Framework pipeline and favorable-regime diagnostics from the manuscript.

05. Weak-Link Quantum Communication

Weak entanglement can become useful when its strength, orientation, and topology are aligned with a task.

RESEARCH PROBLEM

Real networks contain heterogeneous, noisy, and weak inter-group links. Can these links be used as functional resources rather than treated only as defects?

CORE APPROACH

Model cooperating groups as a weak-link network; derive first-order AND/OR corrections; map isotropic and anisotropic noise to effective visibilities; add Bayesian weighting and calibration rules.

MY CONTRIBUTION

Proposed the weak-link task model, sign and basis strategies, analytical corrections, noise mapping, parameter scans, cross-platform feasibility analysis, and simulation workflow.

STATUS First author · Manuscript in preparation

$$\begin{aligned} \mathbb{E}[s_{ij}] &= 2\alpha - 1, \\ \mathbb{E}[\Delta(\mathbf{s})] &= \kappa(2\alpha - 1) \sum_{(i,j)} b_i b_j p_{ij} \end{aligned} \quad (12)$$

By tuning α , one can continuously shift the AND/OR trade-off to suit different task requirements (strong consistency vs. strong fault tolerance).

G. Decoding and Decision: A Bayesian Layer

Under strict no-signaling, decisions can only rely on locally permitted post-processing of each group's measurement outcomes. We adopt a lightweight Bayesian decoding layer: for a given task (AND/OR), a receiver (or an evaluator used for statistical partitioning) forms a likelihood ratio

$$\Lambda(\mathbf{a} | \mathbf{x}) = \frac{P(\mathbf{a} | \text{hypothesis } H_1, \mathbf{x})}{P(\mathbf{a} | \text{hypothesis } H_0, \mathbf{x})}$$

and makes a binary decision at a threshold τ , or performs maximum a posteriori (MAP) decisions in multi-message scenarios. Link heterogeneity (different p_{ij}) and alignment errors (ϵ) enter the likelihood via the statistics induced by Eqs. (8)–(10), thereby yielding decisions consistent with link quality. This decoding layer uses only allowable local statistics and pre-calibrated parameters, requiring no additional communication.

H. Perturbative Characterization of Visibility vs. Capacity/Error

To quantify the “weak-link penalty,” introduce the small parameters $\varepsilon_{ij} := 1 - p_{ij}$. Under appropriate regularity conditions and a fixed measurement/decision architecture, the communication mutual information $I(m; \hat{m})$ (or an equivalent error exponent) can be written as

$$I(m; \hat{m}) \geq I_{\text{EA}}^{(\text{ideal})} - C_1 \varepsilon - C_2 \varepsilon^2 + o(\varepsilon^2) \quad (13)$$

comparison, the parameterization of this chapter will be used in later simulations and experiments under the following protocol:

- Visibility grid:** $p_{ij} \in \{p_{\min} : \Delta p : p_{\max}\}$, with a typical range $[0.75, 0.95]$;
- Alignment factor:** $\kappa \in \{0.80, 0.85, 0.90, 0.95, 1.00\}$;
- Basis strategies:** deterministic optimum s_{ij}^* (Eq. (11)) and randomized mixture $s_{ij} \sim \text{Bernoulli}(\alpha)$ (Eq. (12));
- Statistical evaluation:** for each parameter point, repeat N independent trials to record $\hat{P}_{\text{AND}}$ and $\hat{P}_{\text{OR}}$, and provide confidence intervals (e.g., Clopper-Pearson) with multiple-comparison correction (FDR/Bonferroni) to validate the effective domain of Eqs. (9)–(10);
- Platform mapping:** photonic / trapped-ion / superconducting platforms map hardware-calibrated physical quantities (wave-plate angles, gate fidelities, crosstalk strengths) to (p_{ij}, κ) and generate task curves and phase diagrams under a unified script.

J. Summary

This chapter establishes a multi-group quantum communication framework under the *no-signaling* constraint, targeting **task-level indicators** (AND/OR). The key ingredients are:

- Unifying inter-group resources and noise via ρ_{net} and the correlation tensor T^* ;
- Capturing alignment errors through the geometric factor κ ;
- Linking *weak links – basis strategies – task performance* via the first-order correction Δ (Eq. (8));

A. Analytically characterize the weak-link penalty

TABLE I: Noise-visibility-effective weak-link strength correspondence table (per-link evaluation)

Noise model	Axial visibility (v_x, v_y, v_z)	Basis choice s_{ij}	Effective strength $\rho_{\text{eff}}^{\text{eff}}(s_{ij})$	Substitution
Werner	$v_x = v_y = v_z = p$	arbitrary	p	(9), (3)
Depolarizing	$v_x = v_y = v_z = (1 - q)^2$	arbitrary	$(1 - q)^2$	(9), (3)
Phase-damping	$v_x = v_y = (1 - 2\delta)^2, v_z = 1$	$\pm 1 \rightarrow X, -1 \rightarrow Z$	$\frac{1 + 2\delta_z}{2}(1 - 2\delta)^2 + \frac{1 - \delta_z}{2}$	(9), (3)
Amplitude damping	$v_x = v_y = (1 - \eta), v_z = (1 - \eta)^2$	same as above	$\frac{1 + \delta_z}{2}(1 - \eta) + \frac{1 - 2\delta_z}{2}(1 - \eta)^2$	(9), (3)

^a For each link, measure the visibilities v_x, v_y and the alignment factor δ . Select the optimal basis s_{ij}^* according to the task rule (Eqs. (37)–(38)), compute the effective strength $\rho_{\text{eff}}^{\text{eff}}$, substitute into Eqs. (10)–(14) to obtain the predicted value of $P_{\text{AND/OR}}$. Uncertainty is given by Eqs. (67).

TABLE II: Representative parameter ranges and performance thresholds.

Platform / scenario	κ	Target gain δ	S_0 (experimental estimate)	Isotropic threshold $p \geq \frac{1}{4}$	Predicted gain $\kappa p S_0$
Photonic (SFDC)	0.90	0.05	0.60	$p \geq 0.0025$	0.324
Ion-trap	0.95	0.05	0.50	$p \geq 0.1003$	0.238
Superconducting	0.93	0.05	0.55	$p \geq 0.0082$	0.282

^a The parameter δ is the minimum desired AND/OR gain where the baseline (choose either task); S_0 is inferred from experimental data without inter-group weak links; the column “Predicted gain” gives $\kappa p S_0$ for the specified (κ, p, S_0) . In an actual experiment, these illustrative values should be replaced by the measured S_0 and the platform-calibrated (κ, p) .

3. Only two-qubit states with **partial entanglement** (weak entanglement) must be generated in a controllable way;

3. Measurement operations are standard single-qubit rotations and projective readout.

Hence the scheme is implementable on the three mainstream platforms: photonics, trapped ions, and superconducting circuits.

The physical parameters (ρ, κ) have the following meanings:

ρ = visibility of the entangled link,
 κ = measurement-basis alignment factor. (48)

- ρ is limited by source purity, decoherences, and detection noise.
- κ is governed by basis-setting errors, interferometer phase stability, or gate-calibration accuracy.

The experimental goal is to validate the analytical formulas of Sec. II-IV. The primary observations are:

$P_{\text{AND}}, P_{\text{OR}}, G_{\text{AND/OR}} = \frac{P_{\text{AND/OR}} - P_{\text{AND/OR}}^{\text{rand}}}{P_{\text{AND/OR}}^{\text{rand}}}, \quad (49)$

B. Photonic Implementation

a. *B.1. Extended source and weak-link preparation.* A β -BBO (or PPKTP) crystal pumped at the second-harmonic generates polarization-entangled pairs via SFDC, projecting $|\Psi^{\pm}\rangle = (|HV\rangle \pm |VH\rangle)/\sqrt{2}$. To obtain controllable weak entanglement:

- Insert a polarization depolarizer or liquid-crystal retarder in one arm to realize a Werner mixture:
 $\rho_W(\rho) = \rho |\Psi^{\pm}\rangle\langle\Psi^{\pm}| + \frac{1 - 2\rho}{4} \mathbb{1}_4, \quad \rho \in [0.75, 0.95]$ tunable. (50)
- Alternatively, use a Mach-Zehnder or a Sagnac loop to introduce phase mismatches and emulate phase-damping behavior.

Citation note: Our standard SFDC entangled-source implementations and Sagnac/MZL entanglement sources, e.g., [7, 77].

b. *B.2. Measurement and basis alignment.* Single-qubit bases are set by HW/FQWP stacks. A controlled angular offset θ realizes an effective alignment factor $\kappa = \cos(2\theta)$; e.g., $\theta = 10^\circ$ gives $\kappa \approx 0.94$.

SELECTED EVIDENCE

- The same variables support coordination-oriented AND and redundancy-oriented OR tasks.
- Four representative noise models are expressed through a common visibility framework.
- Photonic, trapped-ion, and superconducting mappings connect theory with tunable parameters.

NEXT QUESTION Representing weak links as weighted edges raised a system-level question: how should nodes adapt when topology, noise, cooperation, and competition change together?

Analytical task curves, robustness diagnostics, and platform mapping.

06. Learnable Weak-Link Multi-Group Quantum Games

Analytical structure can constrain learning while trainable variables adapt to topology and noise.

RESEARCH PROBLEM

Can a weak-link quantum network become an adaptive decision system rather than a passive communication medium?

CORE APPROACH

Represent groups as vertices and weak resources as signed weighted edges; derive a game tensor, sign rules, advantage criteria, and spectral bounds; relax discrete structure into differentiable variables.

MY CONTRIBUTION

Proposed the unified analytical-learning framework; derived first-order and spectral structure; implemented automated search, repeated trials, batch comparisons, visualization, backend evaluation, and cloud-hardware checks.

STATUS First author · Submitted to Quantum Machine Intelligence · Under editorial review

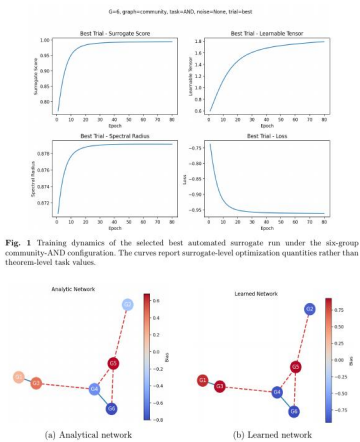


Fig. 1 Training dynamics of the selected best automated surrogate run under the six-group community-AND configuration. The curves report surrogate-level optimization quantities rather than theorem-level task values.

Fig. 2 Comparison between the analytical sign rule and the learned network structure for the selected six-group community-AND surrogate run.

7.3 Batch comparison across topology, task, and noise

The automated search result is complemented by a newly generated explicit batch protocol over graph type, task type, group count, and noise setting. Table 3 summarizes the highest-performing batch configurations after sorting by the final surrogate score. The strongest explicit batch configurations reach $P_{\text{win}}^{\text{net}} \approx 0.998$ under the current surrogate protocol. Because the updated batch is a single-seed implementation-side

SELECTED EVIDENCE

- Selected surrogate runs remain aligned with analytical sign rules while adapting continuous parameters.
- Topology, task structure, and noise are evaluated through reproducible experiments.
- The work separates theorem-level results, surrogate objectives, and backend proxy quantities.

NEXT QUESTION This motivates quantum machine intelligence for future networks: learning should allocate links, measurements, routing, and redundancy while preserving interpretability.

Research Synthesis

One scientific question, four complementary answers

01 · SECURITY

Protocol structure determines whether a quantum primitive retains its security role.

Reliable advantage begins with authenticated information flow, protected verification authority, and composable security.

02 · RESOURCE ALLOCATION

Raw success must be normalized by logical and physical overhead.

Support geometry, conditional discrimination, and convex certificates identify when finite resources are genuinely useful.

03 · NETWORK FUNCTION

Weak links can be designed around a task rather than simply removed.

Visibility, basis alignment, topology, and noise become tunable variables in distributed quantum communication.

04 · ADAPTIVE DECISION

Learning can make network structure responsive without replacing analytical reasoning.

Differentiable models should operate inside physically meaningful and mathematically interpretable constraints.

I do not regard imperfect resources as obstacles to be removed before useful computation begins. I regard them as structures that can be modeled, allocated, certified, and exploited.

Methods and Technical Toolkit

Theoretical modeling, computational validation, and reproducible implementation

THEORY & MODELING

- Quantum protocol and information-flow analysis
- Graph-based weak-link network models
- POVM-based state discrimination
- Resource-normalized performance criteria
- First-order perturbative analysis
- Spectral and scalability bounds

OPTIMIZATION & COMPUTATION

- Python scientific computing and numerical experiments
- Qiskit and PennyLane circuit simulation
- Semidefinite programming and certificate logic
- Automated search and batch evaluation
- Noise, robustness, and sensitivity analysis
- Visualization and statistical summaries

RESEARCH ENGINEERING

- Reproducible experiment pipelines
- Artifact persistence and structured outputs
- LaTeX manuscript and appendix preparation
- Figures, tables, references, and notation consistency
- Git/Linux/Ubuntu development workflows
- Full-stack software and server deployment

SELECTED EVIDENCE OF PREPARATION

PREPARATION

Top 10% in Software Engineering; preparation in algorithms, data structures, linear algebra, probability, databases, and programming.

PREPARATION

Provincial mathematics and modeling awards; experience translating physical and engineering questions into computational models.

PREPARATION

Independent software development involving backend systems, web interfaces, Electron clients, and Ubuntu deployment.

Future Research Agenda

Toward a resource-aware, adaptive, and interpretable Quantum Internet

RESOURCE-AWARE NETWORK PROTOCOLS

Jointly optimize topology, routing, entanglement quality, security, and task performance under finite-resource constraints.

- Heterogeneous links
- Adaptive entanglement allocation
- End-to-end task metrics
- Resource-aware security

LEARNABLE QUANTUM NETWORKS

Develop graph-structured learning systems that adapt measurements and policies under calibrated noise while preserving analytical guarantees.

- Graph learning
- Variational optimization
- Bayesian decisions
- Convex certification

UNIFIED DISTRIBUTED ARCHITECTURES

Study communication, computation, cryptography, and error correction as interacting components consuming the same limited resources.

- Distributed quantum computing
- Quantum cryptography
- Error-aware orchestration
- Interpretable control

To become an independent researcher developing intelligent and resource-efficient quantum information systems that integrate communication, computation, security, and learning.

Selected Outputs and Contact

Current research record and application profile

MANUSCRIPTS AND PAPERS

On Grover-Based Quantum Signature Schemes

Second author · Revised and resubmitted

A Learnable Unified Framework for Weak-Link Multi-Group Quantum Games

First author · Under editorial review

Weak-Link Entanglement-Assisted No-Signaling Quantum Communication Protocols

First author · Manuscript in preparation

Error-Corrected Quantum-Huffman-Inspired Probabilistic Source Decoding

First author · Manuscript in preparation

HONORS

- Two Provincial Third Prizes, Chinese Mathematics Competition (2024, 2025).
- Provincial Third Prize, Contemporary Undergraduate Mathematical Contest in Modeling (2025).
- Honorable Mention, Shuwei Cup Mathematical Modeling Challenge (2026).

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Fall 2027

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RESEARCH FOCUS

Quantum Information · Communication · Cryptography · Networks

Quantum Machine Intelligence